

1. Determine the points of maximum and minimum of the function

$$f(x, y) := \int_0^{x+y^2} e^{-t^2} dt \quad \text{subject to} \quad 2x^2 + y^2 \leq 1.$$

2. **Evaluation of double integrals over rectangular domains:**

Let $D := [a, b] \times [c, d]$ and $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function. Then the double integral of the function f over the domain D is evaluated as follows:

$$\iint_D f(x, y) \, dx dy = \int_a^b \left(\int_c^d f(x, y) \, dy \right) dx = \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy.$$

Evaluate the double integrals:

- (a) $\iint_D \sin(x - y) \, dx dy$ where $D := [0, \frac{\pi}{2}] \times [0, \pi]$.
- (b) $\iint_D x \arctan\left(\frac{1}{x}\right) \cos^3(y) \, dx dy$ where $D := [1, \sqrt{3}] \times [0, \frac{\pi}{2}]$.
- (c) $\iint_D \cos^2(x + y) \, dx dy$ where $D := [0, \frac{\pi}{4}] \times [-\frac{\pi}{4}, \pi]$.

$$\text{Hints: } \begin{cases} \cos^3(\theta) = (\cos^2(\theta)) \cos \theta = (1 - \sin^2(\theta)) \cos \theta = \cos \theta - \sin^2(\theta) \cos \theta. \\ \cos(2\theta) = 2 \cos^2(\theta) - 1. \end{cases}$$

3. **(To do after the lecture on Friday)** Let R be the region in the first quadrant bounded by the graphs of the parabolas $y = 2 - x^2$, $y = x^2$ and the y -axis.

- (i) Find the points of intersection of the parabolas.
- (ii) Sketch the region R .
- (iii) Express the area of the region R as a repeated integral, integration first with respect to y , and then with respect to x .
- (iv) Express the area of the region R as a repeated integral, integration first with respect to x , and then with respect to y .
- (v) Use either (iii) or (iv) to calculate the area of R .
- (vi) Let R' be the region bounded by the parabolas $y = 2 - x^2$ and $y = x^2$. Find the area of R' .

4. (To do after the lecture on Friday)

(a) (i) Can you integrate $\sin(x^2)$ with respect to x ?

(ii) Evaluate $\int_0^{\pi^{1/4}} \int_{y^2}^{\sqrt{\pi}} y \sin(x^2) dx dy$.

(Hint: Sketch the domain of integration)

(b) Evaluate the double integral

$$\iint_D \frac{x}{(x+y)^2} dx dy \text{ where } D := \{(x, y) \in \mathbb{R}^2: 1 \leq x \leq 2, \quad x^2 \leq y \leq 2x^2\}.$$